

Theoretical Neuroscience : Chapter 5

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HYKE Seminar

Overview

- The Hodgkin-Huxley Model
- Modeling Channels
- Synaptic Conductances

Recall: Single-compartment model

$$c_m \frac{dV}{dt} = -i_m + \frac{I_e}{A}$$

- V is membrane potential.
- c_m is membrane capacitance.
- i_m is a current per unit area.
- I_e is electrode current.
- A is total surface area of the neuron.

The Hodgkin-Huxley Model

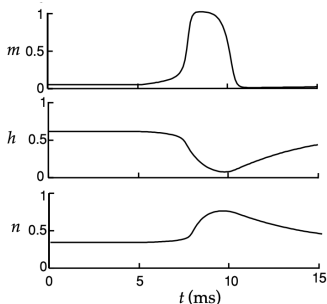
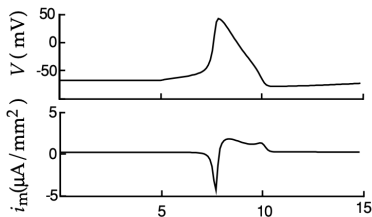
In single-compartment form:

$$\begin{cases} c_m \frac{dV}{dt} = -i_m + \frac{I_e}{A}, \\ i_m = \bar{g}_L(V - E_L) + \bar{g}_K n^4 (V - E_K) + \bar{g}_{Na} m^3 h (V - E_{Na}), \\ \frac{df}{dt} = \alpha_f(V)(1 - f) - \beta_f(V)f, \quad f = n, m, h \end{cases}$$

- leakage current, delayed-rectified K^+ current, transient Na^+ current
- α_f : opening rate, β_f : closing rate
- n : K^+ channels, m , h : Na^+ channels

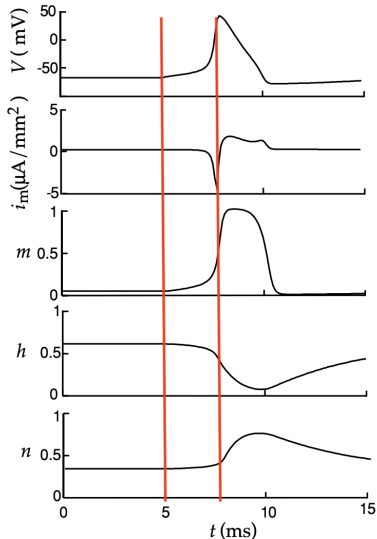
The Hodgkin-Huxley Model

$$\begin{cases} c_m \frac{dV}{dt} = -i_m + \frac{I_e}{A}, \\ i_m = \bar{g}_L(V - E_L) + \bar{g}_K n^4(V - E_K) + \bar{g}_{Na} m^3 h(V - E_{Na}), \\ \frac{df}{dt} = \alpha_f(V)(1 - f) - \beta_f(V)f, \quad f = n, m, h \end{cases}$$



$I_e = 0$ until $t = 5$, $I_e > 0$ for $t > 5$

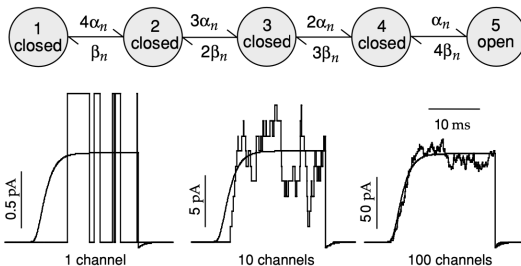
The Hodgkin-Huxley Model



- $t = 5$, dynamics start
- V increases
- Na^+ channels activated (m increases, h decreases)
- sudden drop of i_m
- K^+ channels activated (n increases)
- i_m increases, V decreases

Modeling Channels

- one single channel = multiple subunit gates



Example 1. K^+ channel in the Hodgkin-Huxley model with 4 subunit gates, state i : $i - 1$ subunit gates are open ($i = 1, \dots, 5$)

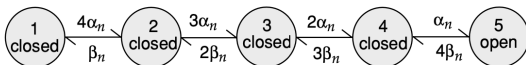
Modeling Channels

Recall the Hodgkin-Huxley model

$$\begin{cases} c_m \frac{dV}{dt} = -i_m + \frac{I_e}{A}, \\ i_m = \bar{g}_L(V - E_L) + \bar{g}_K n^4 (V - E_K) + \bar{g}_{Na} m^3 h (V - E_{Na}), \\ \frac{df}{dt} = \alpha_f(V)(1 - f) - \beta_f(V)f, \quad f = n, m, h, \end{cases}$$

$\implies n$: probability of one subunit gate opens

Modeling Channels



p_i : probability of K^+ channel is in state i ($i = 1, \dots, 5$)

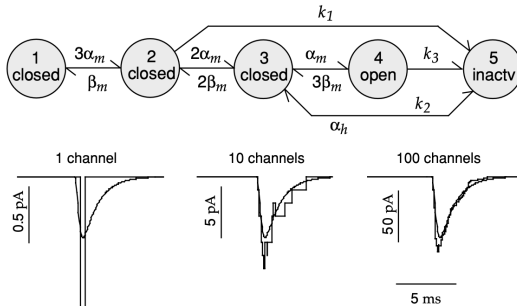
$$p_i = \binom{4}{i-1} n^{i-1} (1-n)^{5-i}, \quad \frac{dn}{dt} = \alpha_n(1-n) - \beta_n n$$

$$\Rightarrow \begin{cases} \dot{p}_1 = \beta_n p_2 - 4\alpha_n p_1, \\ \dot{p}_2 = 4\alpha_n p_1 + 2\beta_n p_3 - (\beta_n + 3\alpha_n) p_2, \\ \dot{p}_3 = 3\alpha_n p_2 + 3\beta_n p_4 - (2\beta_n + 2\alpha_n) p_3, \\ \dot{p}_4 = 2\alpha_n p_3 + 4\beta_n p_5 - (3\beta_n + \alpha_n) p_4, \\ \dot{p}_5 = \alpha_n p_4 - 4\beta_n p_5 \end{cases}$$

Noet that $\sum_{i=1}^5 p_i$ is constant.

Modeling Channels

Recall $i_m = \bar{g}_L(V - E_L) + \bar{g}_K n^4(V - E_K) + \bar{g}_{Na} m^3 h(V - E_{Na})$



Example 2. Na⁺ channel in the Hodgkin-Huxley model

Synaptic Conductances



- $g_s = \bar{g}_s P$: synaptic conductance
- $\bar{g}_s$: maximal conductance
- $P = P_s P_{\text{rel}}$: open channel probability
- P_s : probability that postsynaptic channel opens
- P_{rel} : probability of transmitter from the presynaptic terminal

Synaptic Conductances

$$\frac{dP_s}{dt} = \alpha_s(1 - P_s) - \beta_s P_s$$

- α_s : opening rate, β_s : closing rate
- $\alpha_s \gg \beta_s$ when transmitter concentrated on postsynaptic channel

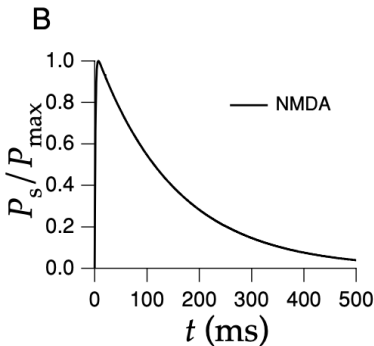
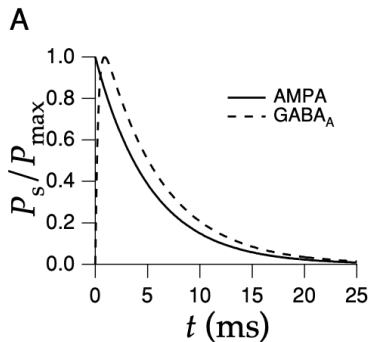
$$\frac{dP_s}{dt} \approx \alpha_s(1 - P_s) \implies P_s(t) = 1 + (P_s(0) - 1)e^{-\alpha_s t}, \quad 0 \leq t \leq T$$

- $\alpha_s = 0$ after $t = T$

$$P_s(t) = P_s(T)e^{-\beta_s(t-T)}, \quad t \geq T$$

Synaptic Conductances

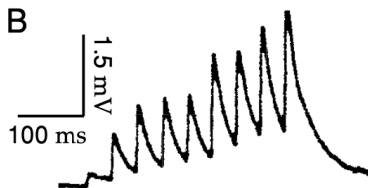
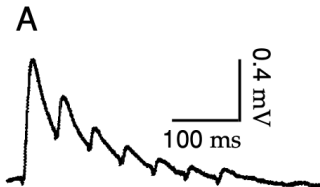
$$\frac{dP_s}{dt} = \alpha_s(1 - P_s) - \beta_s P_s$$



Synaptic Conductances

$$\tau_p \frac{dP_{\text{rel}}}{dt} = P_0 - P_{\text{rel}}, \quad \begin{cases} A \text{ (depression)} & P_{\text{rel}} \rightarrow f_D P_{\text{rel}}, \\ B \text{ (facilitation)} & P_{\text{rel}} \rightarrow P_{\text{rel}} + f_F(1 - P_{\text{rel}}), \end{cases}$$

$$(0 \leq f_F, f_D \leq 1)$$



Synaptic Conductances

$\langle P_{\text{rel}} \rangle$: average steady-state release probability

$$\tau_p \frac{dP_{\text{rel}}}{dt} = P_0 - P_{\text{rel}}, \quad P_{\text{rel}} \rightarrow P_{\text{rel}} + f_F(1 - P_{\text{rel}}).$$

Assume spike happens every τ

$$\begin{aligned} P_{\text{rel}}(0+) &= \langle P_{\text{rel}} \rangle + f_F(1 - \langle P_{\text{rel}} \rangle), \\ P_{\text{rel}}(\tau) &= P_0 + [\langle P_{\text{rel}} \rangle + f_F(1 - \langle P_{\text{rel}} \rangle) - P_0] e^{-\tau/\tau_p}. \end{aligned}$$

Assume $\tau \sim re^{-r\tau}$

$$\begin{aligned} \langle P_{\text{rel}} \rangle &= P_0 + [\langle P_{\text{rel}} \rangle + f_F(1 - \langle P_{\text{rel}} \rangle) - P_0] \int_0^\infty re^{-r\tau} e^{-\tau/\tau_p} d\tau \\ &= P_0 + [\langle P_{\text{rel}} \rangle + f_F(1 - \langle P_{\text{rel}} \rangle) - P_0] \frac{r\tau_p}{1 + r\tau_p}. \end{aligned}$$

Synaptic Conductances

$\langle P_{\text{rel}} \rangle$: average steady-state release probability

$$\langle P_{\text{rel}} \rangle = \begin{cases} \text{(depression)} & \frac{P_0}{1 + (1 - f_D)r\tau_p} \\ \text{(facilitation)} & \frac{P_0 + f_F r \tau_p}{1 + f_F r \tau_p} \end{cases}$$

Thank you for listening!